CONTINUAL REASONING GYM: Diagnosing and Harnessing Shared Reasoning in Continual RLVR

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https://crg-rl.github.io/ · arXiv:2608.18574 · github.com/crg-rl/crg

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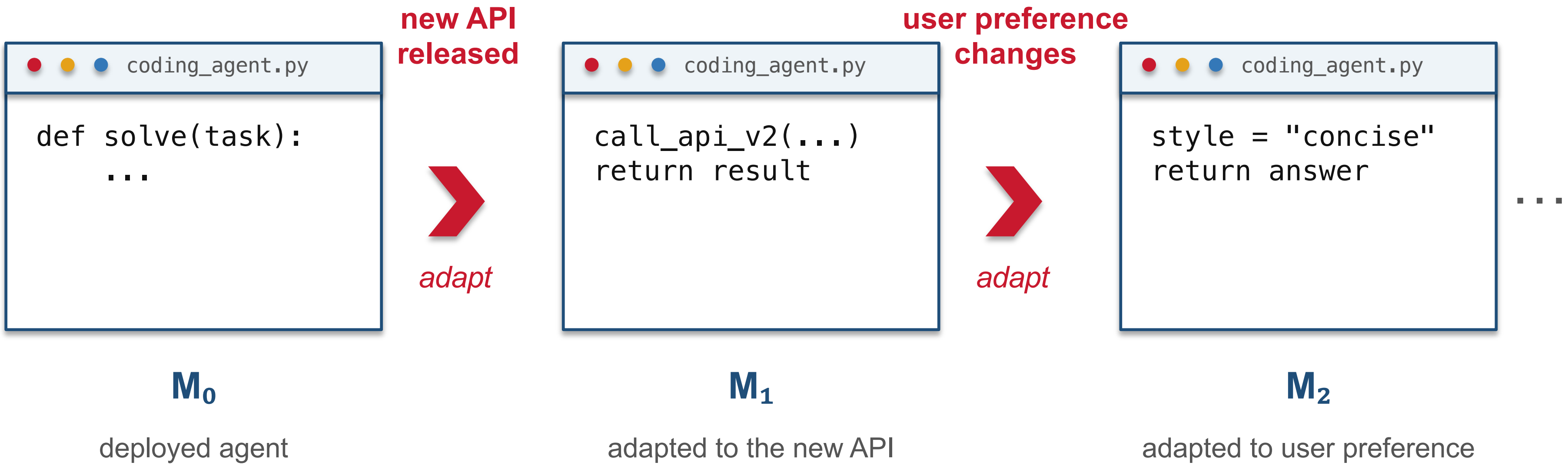
Seq. RLVR forgets modestly yet reaches only **0.88× MTRL**. CPR reaches **1.03×**.

5 task sequences · 30 stages · text and visual reasoning

A deployed coding agent must keep adapting

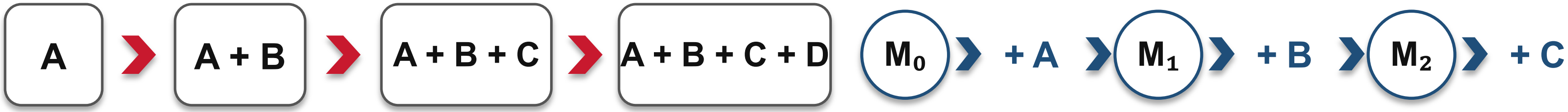


New capability requirements keep arriving after deployment.

Joint retraining becomes costly as capabilities grow

Joint retraining after every task arrival

Task-by-task update



retrain the enlarged mixture at every step

reuse the existing policy

Continual RLVR aims to match joint-training performance

through task-by-task updates.

Can task-by-task RLVR match joint training?

Continual RLVR

$$\tau_1 \rightarrow \tau_2 \rightarrow \tau_3 \rightarrow \tau_T \rightarrow M_T$$

tasks arrive one by one

Multitask RLVR (MTRL)

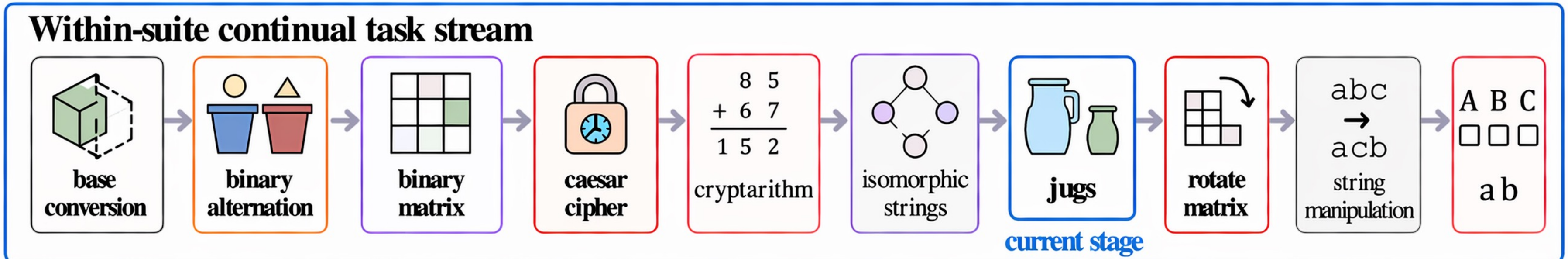
$$\tau_1 \quad \tau_2 \quad \ldots \quad \tau_T \rightarrow M^*$$

joint optimization

$$\textbf{CTM} = \frac{\mathrm{FinalAvg}(M_T)}{\mathrm{FinalAvg}(M^*)} \rightarrow 1$$

CTM compares the continual endpoint with its joint-training reference.

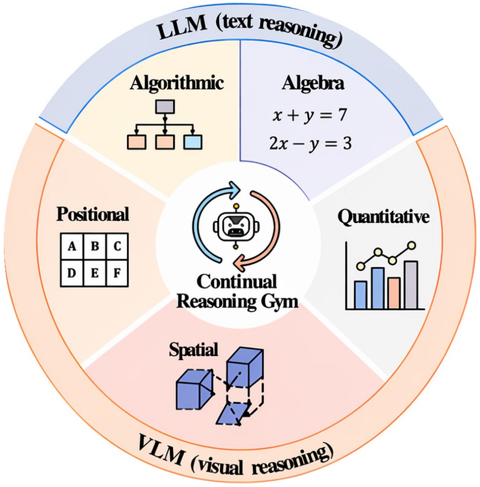
CRG instantiates five continual RLVR streams



At each stage

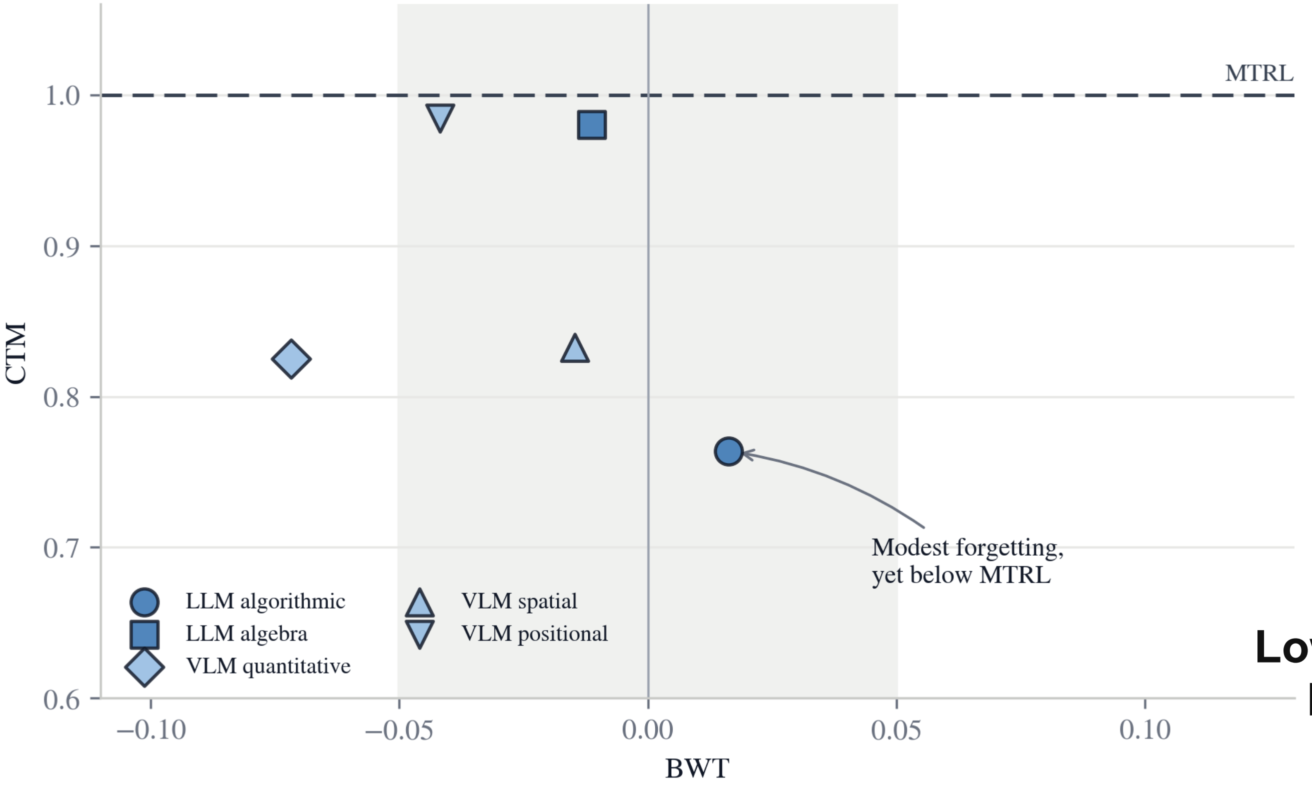
prompt → roll out → verify → update

then move to the next task stage



30 stages across two text-reasoning and three visual-reasoning sequences.

Seq. RLVR forgets modestly yet reaches only 88% of MTRL



−2.47 pp

mean backward transfer

0.88

mean CTM

Low forgetting does not imply MTRL-level performance.

Forgetting explains only part of the MTRL gap

$$\mathbf{FinalAvg} = \mathrm{BaseAvg} + \frac{T-1}{T}\,\mathbf{FWT} + \mathbf{TLG} + \frac{T-1}{T}\,\mathbf{BWT}$$

Initial ability
BaseAvg

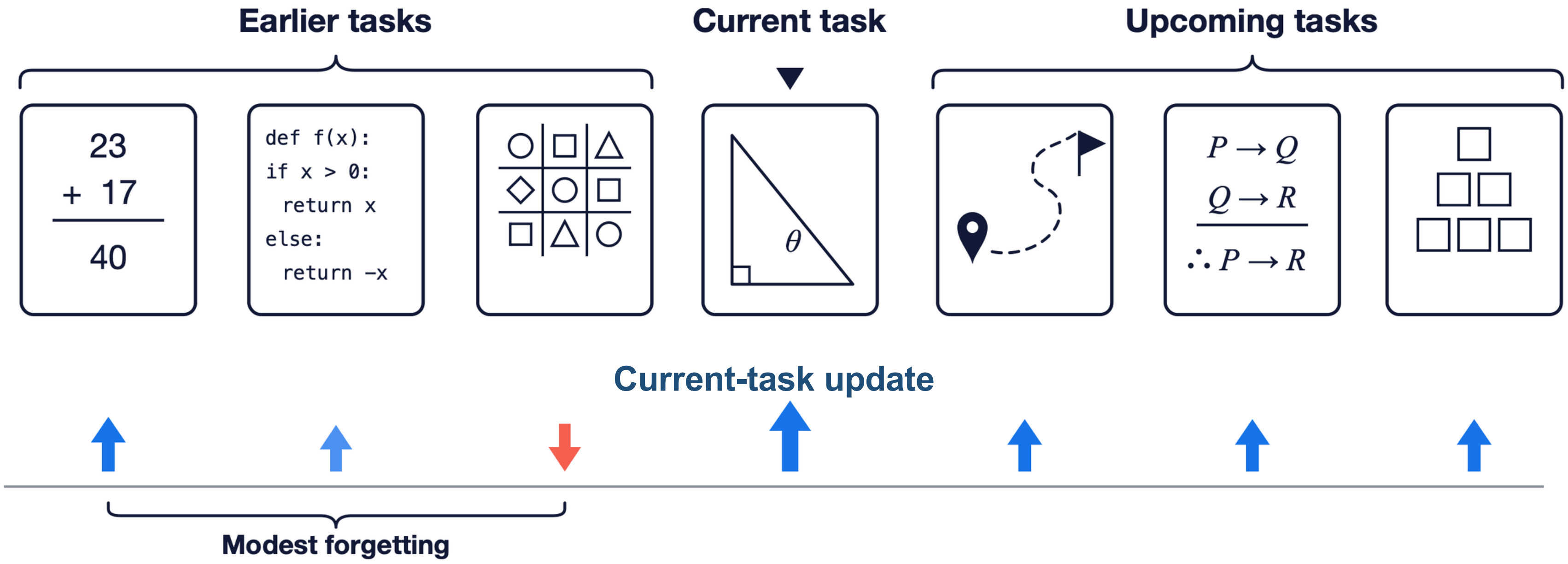
Before its own stage
FWT

During its own stage
TLG

After its own stage
BWT

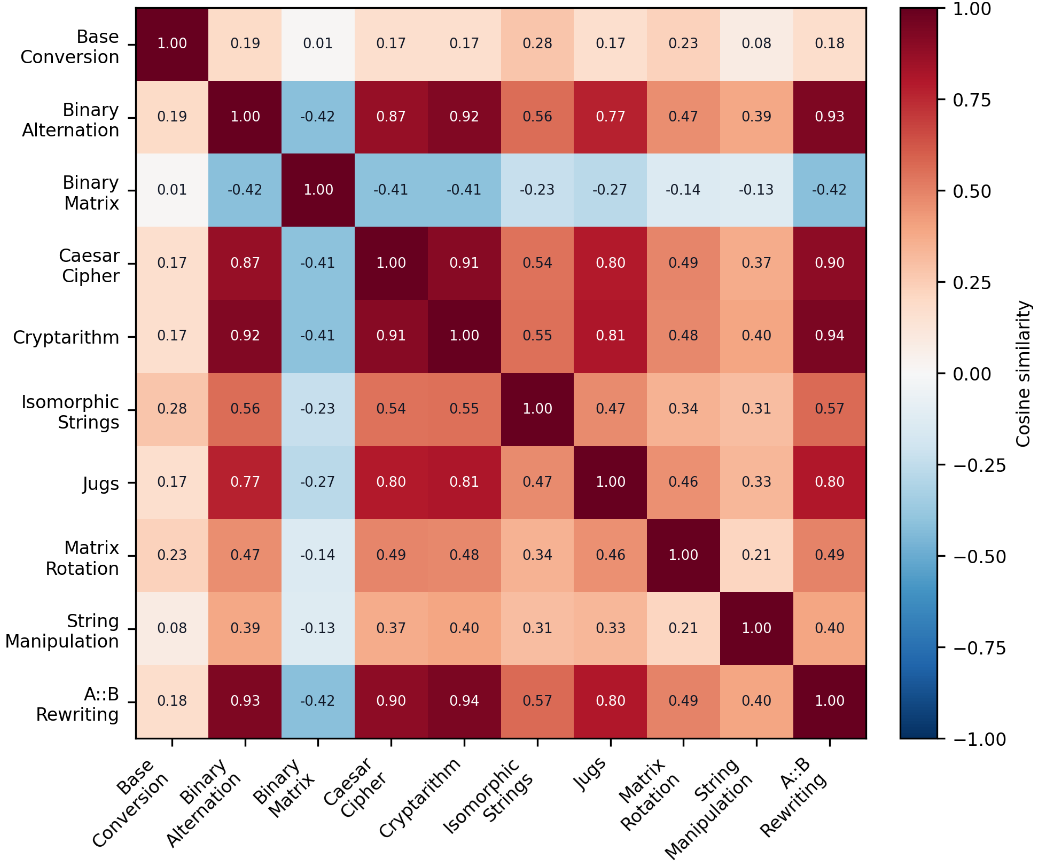
Even eliminating forgetting leaves gaps in **forward transfer** and **direct task learning**.

Shared reasoning explains modest forgetting



Shared reasoning: an update on one reasoning task benefits others on average.
Later-task updates therefore limit net forgetting.

Task gradients are positively aligned on average



0.345

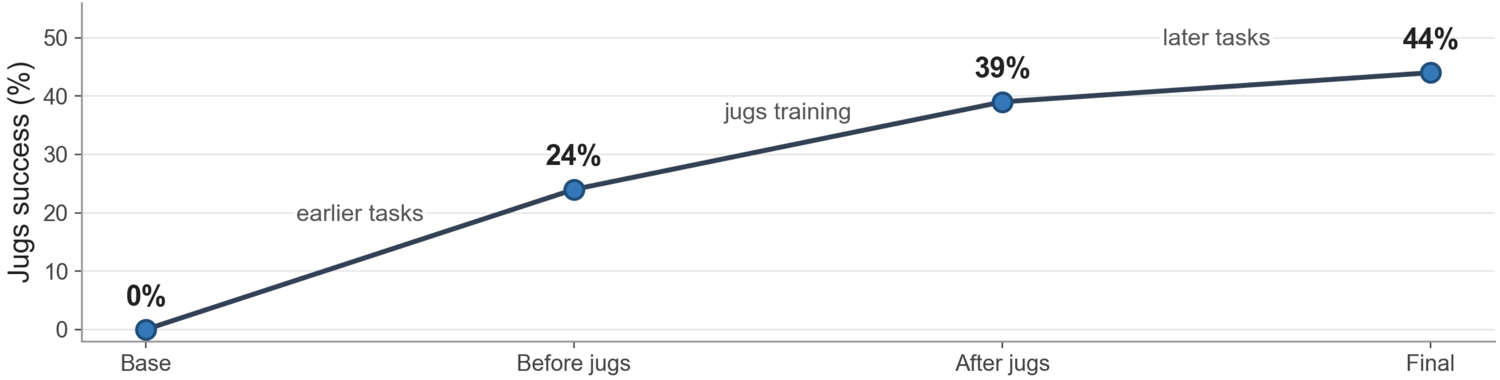
mean pairwise cosine

82.2%

positive task pairs

The positive mean is consistent with shared reasoning.

Shared reasoning also appears in model behavior



Base

"Maybe fill B or C … Wait … There we go! … But let me—"

Second-guesses a valid route

No answer

Before jugs

"Maybe use the 11L jug … that's not getting me to 3. Let me try another approach …"

Finds a longer route

No answer

After jugs

"I have a valid route … let me track each step … B has 3."

Rechecks the route

Valid answer

Final

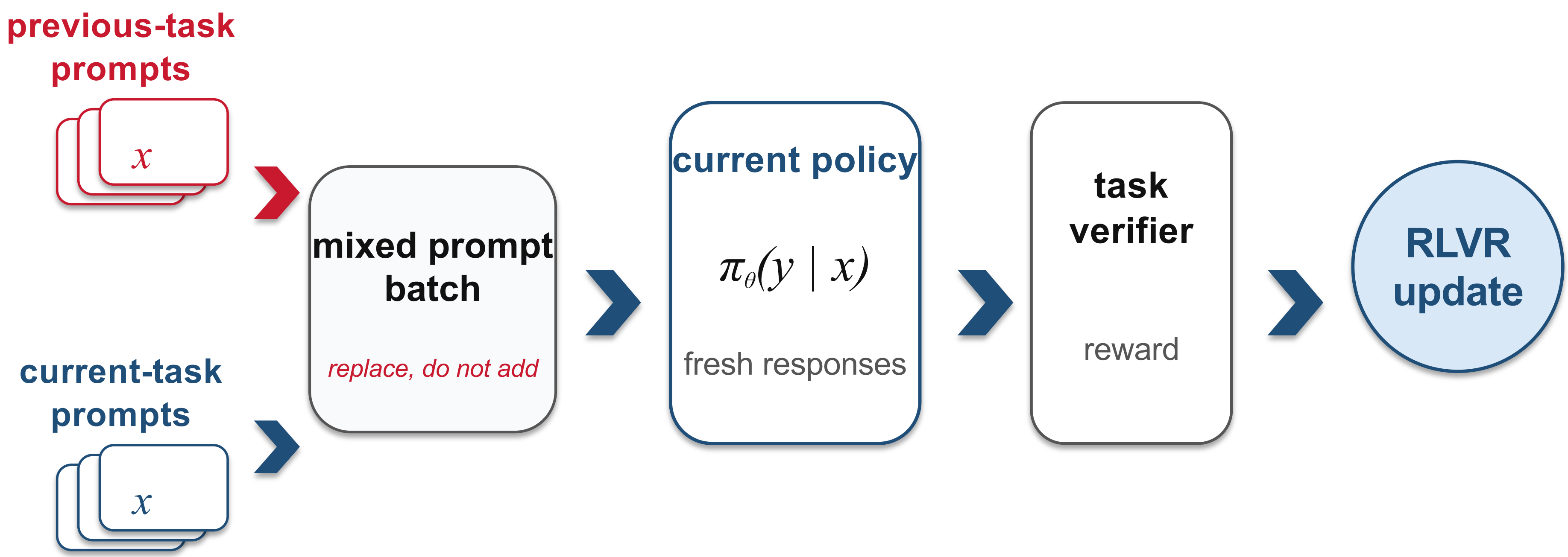
"Fill B. Pour into A … empty A … fill B again … B has 3."

Direct reasoning

Valid answer

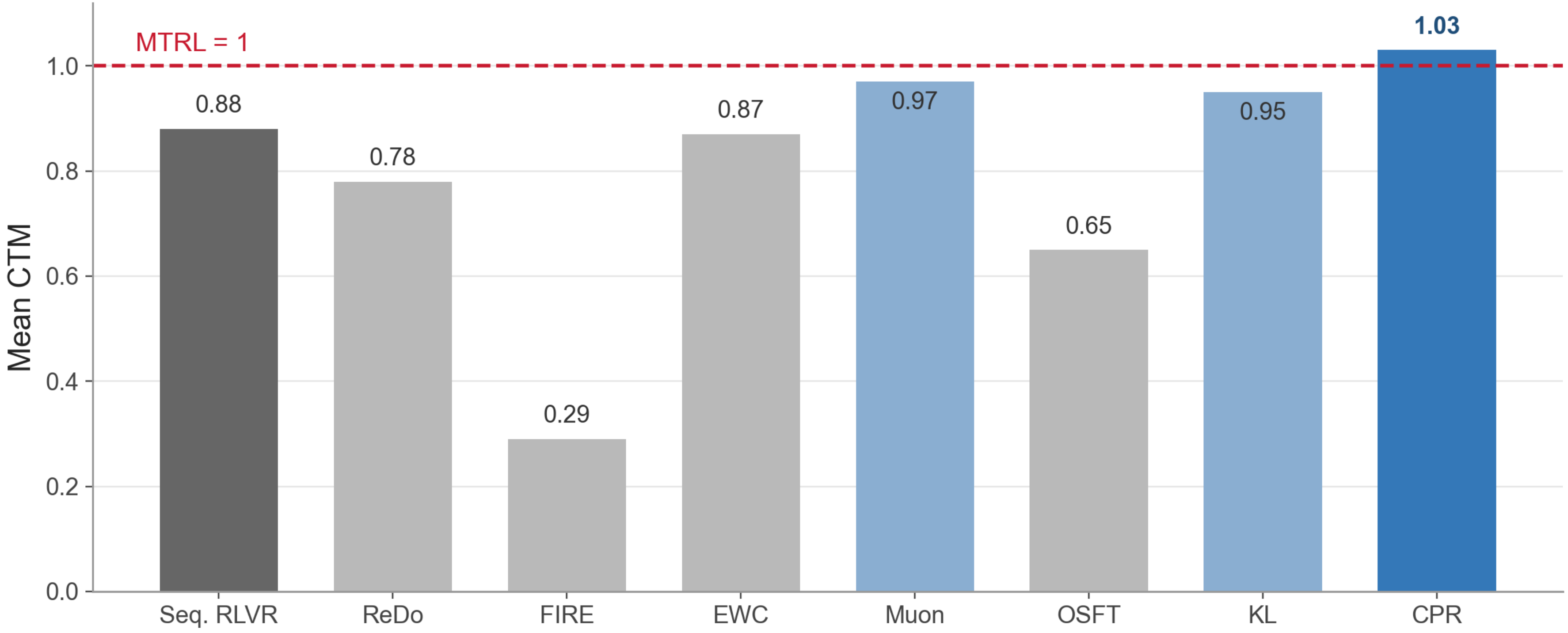
Earlier- and later-task training improve both success and reasoning.

CPR brings previous tasks into current-policy training



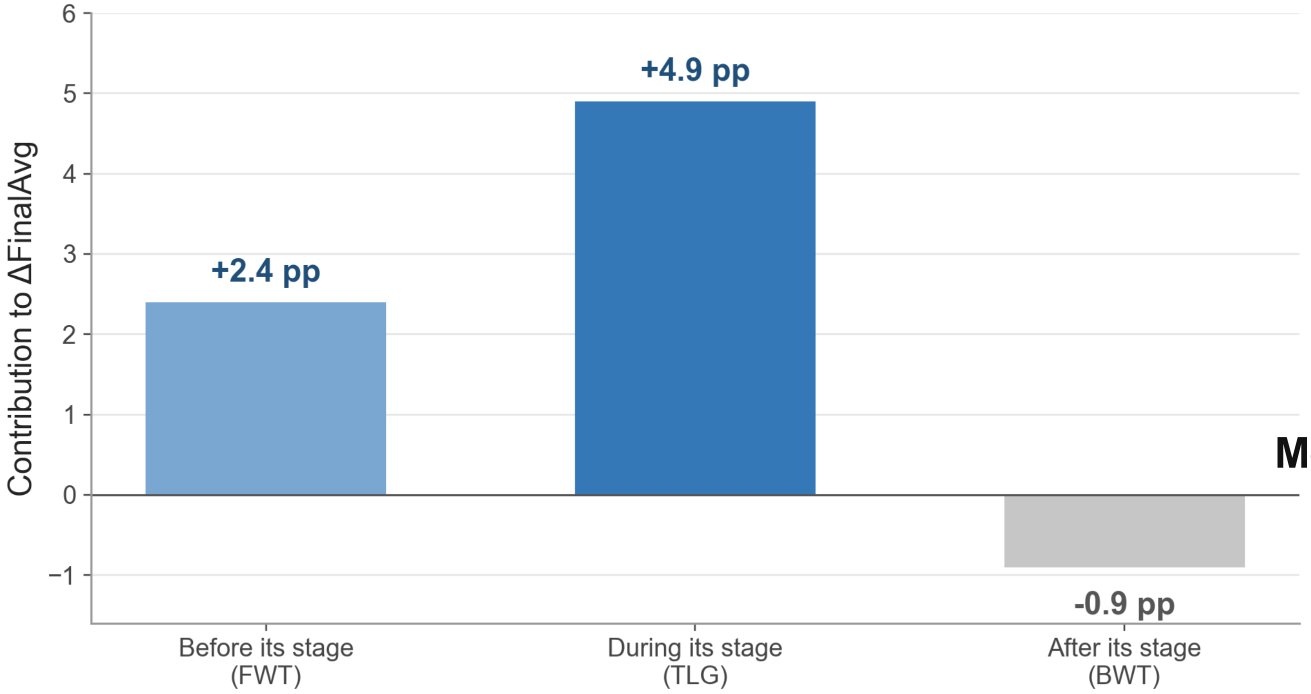
CPR changes the prompt distribution **without increasing prompt or rollout counts**.

Only CPR reaches MTRL-level performance on average



CPR reaches 1.03 mean CTM, compared with 0.88 for Seq. RLVR.

CPR improves learning on arriving and future tasks

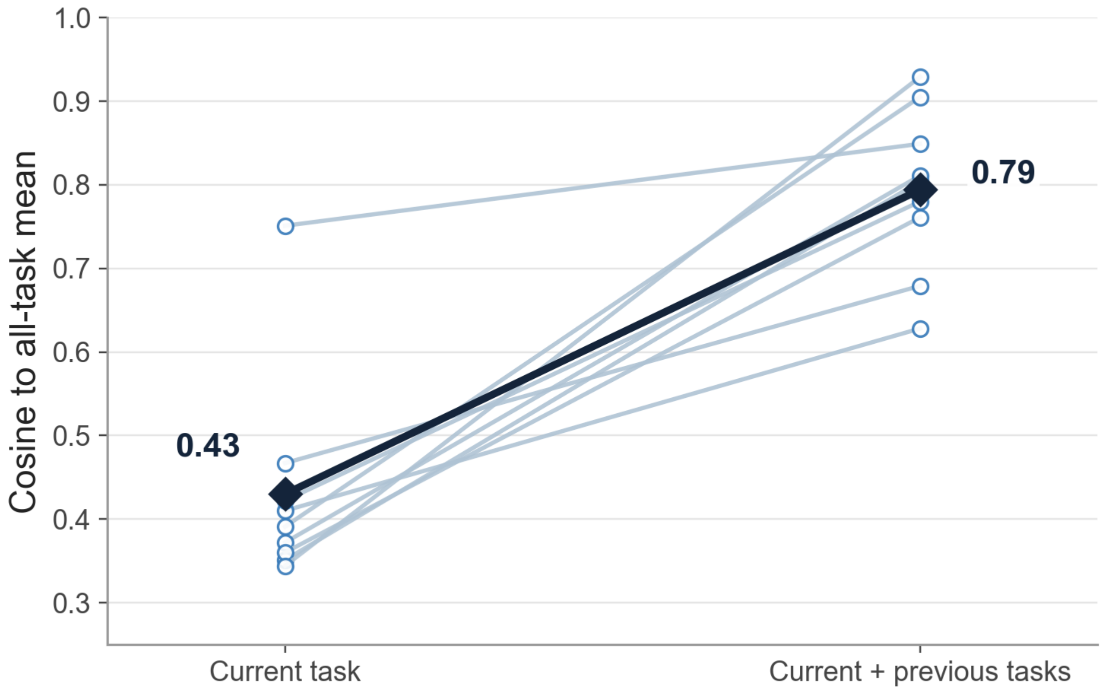


+6.4 pp

mean FinalAvg gain

Most of the gain comes from direct task learning and forward transfer.

Previous-task gradients improve alignment with the full task set



0.43 → 0.79

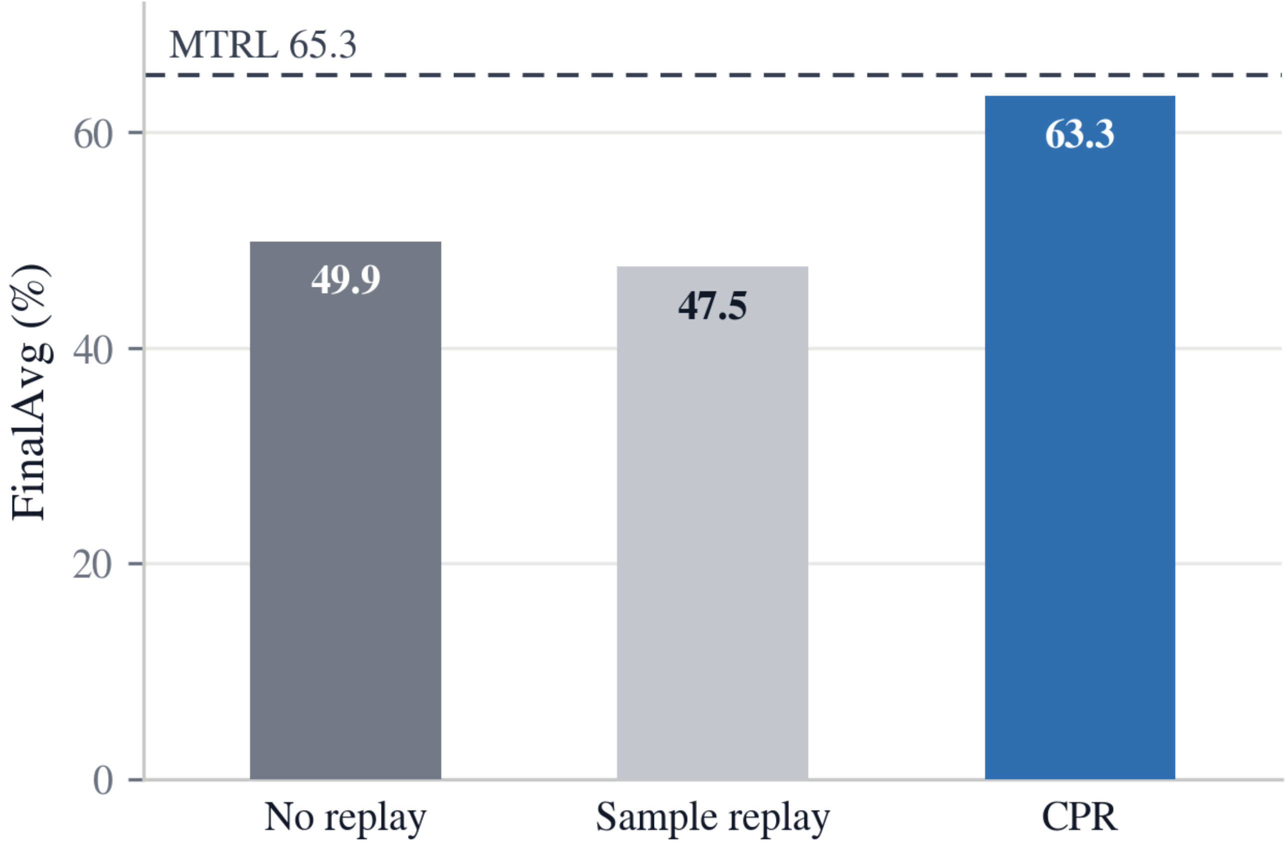
cosine to the all-task mean

9 / 9

stage comparisons improve

The mixed direction is closer to the gradient of the full task set.

Current-policy regeneration is critical



63.3

CPR FinalAvg

47.5

sample replay

Replay prompts.
Regenerate responses.

Takeaways

- **Seq. RLVR** forgets modestly, yet reaches only 0.88 mean CTM.
- **Shared reasoning** explains modest forgetting and motivates CPR.
- **CPR** harnesses shared reasoning and reaches 1.03 mean CTM.

Project: https://crg-rl.github.io/ **Code:** github.com/crg-rl/crg